SAM ADEYEMI

Backend, Web & AI Software Engineer · Python / FastAPI · PHP / Drupal · PostgreSQL · LLMs, RAG & Agentic AI

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PROFESSIONAL PROFILE

Backend and AI software engineer with 6+ years building secure, scalable backend services and shipping LLM-powered features to production, not just prototypes. Hands-on across the full backend: REST APIs and third-party integrations, PostgreSQL schema design and query optimisation on large event datasets, background jobs, real-time features, role-based access control and audit logging. On the AI side: OpenAI and Anthropic integration, structured outputs, RAG with embeddings and vector search, and multi-agent orchestration. Deploys self-hosted client software on Linux with Docker and Nginx, backed by automated testing and CI. On the web side: around 15 years building WordPress sites, and a Drupal 11 council-services project with a custom module, an accessible WCAG 2.2 AA theme and automated accessibility checks. MSc Information Security & Software Engineering (Distinction), First Class BEng Software Engineering, peer-reviewed AI governance research, and a C#/ASP.NET and SQL Server foundation from five academic projects.

CORE TECHNICAL SKILLS

Languages	Python, SQL, PHP, TypeScript / JavaScript, C# (academic)
Backend & APIs	FastAPI, ASP.NET (academic), REST API design, third-party API integration, background job processing, real-time features, event-driven workflows
Data	PostgreSQL, Microsoft SQL Server, MySQL, pgvector; schema design, indexing and query optimisation on large event datasets
AI & LLM	OpenAI and Anthropic APIs: prompting, structured outputs, context handling; RAG, embeddings, vector and semantic search; agentic and multi-agent orchestration
AI-assisted dev	Claude Code and GitHub Copilot in daily development
Security	Role-based access control, audit logging, API key management, secure-by-design APIs; AI governance (published research)
DevOps & quality	Docker, Nginx reverse proxy, Linux servers, self-hosted client deployments, unit and integration testing, CI pipelines, Git / GitHub
Web & CMS	Drupal 11 (custom modules, services, Form API, Twig theming), WordPress (around 15 years)
Accessibility	WCAG 2.2 AA, GOV.UK Design System patterns, automated checks with axe-core and Playwright
Frontend	React with TypeScript, Tailwind CSS, shadcn/ui

PROFESSIONAL EXPERIENCE

Lead Software & AI Systems Engineer | Apex Digital Africa **2019 – Present**

Web development and AI consultancy delivering backend, SaaS and AI products for clients and in-house platforms

- Design, build and maintain secure backend services, **REST APIs and third-party integrations** in Python and FastAPI, from requirements through testing, release and production support.
- Ship **LLM-powered features to production** using OpenAI and Claude APIs, with prompt design, structured outputs and context handling for domain-specific retrieval and workflow automation.
- Re-engineered project workflows around **event-driven and metadata-driven design** with schema-validated outputs, achieving **85% faster execution** and **73% fewer output errors**.
- Design PostgreSQL schemas, **indexes and queries for large event datasets**, and build **background jobs and real-time features** that keep dashboards and reports responsive.
- Implement **role-based access control, audit trails and reporting** for multi-user SaaS platforms, so every action is attributable.
- Deploy and run client software on **self-hosted Linux servers with Docker and Nginx**, backed by automated unit and integration tests and CI pipelines.

- Build React and TypeScript front ends with Tailwind CSS and shadcn/ui; explain technical trade-offs to non-technical clients without jargon.

SELECTED PROJECTS

Council Services on Drupal 11 *github.com/SamAde1203/council-services-drupal*

Custom Drupal 11 module and accessible theme for a fictional council: residents enter a postcode to find their bin days, verified against the postcodes.io API with caching, rate limiting and a fallback if the API is down. 26 PHPUnit tests, Drupal coding standards and WCAG 2.2 AA checks (0 axe-core violations) run in GitHub Actions.

PHP 8.3 · Drupal 11 · Twig · PHPUnit · Playwright / axe-core · GitHub Actions · DDEV

PHHM Multi-Agent AI Platform *Lead Architect & Developer*

Architected an agentic platform where a central orchestrator routes work to seven specialist agents for communication, analytics, knowledge retrieval and user interaction, serving users across multiple countries.

Python · FastAPI · OpenAI API · REST · JSON / YAML · Linux

MR_CLAUDE_NOTEBOOK_LM: AI Engineering Operating System *In development*

Retrieval pipeline over technical documents (chunking, embeddings, vector storage and semantic search) behind a custom FastAPI backend, used for documentation analysis and API security review. Answers are validated against retrieved evidence; scenarios include API-key rejection and audit-event reasoning.

Python · FastAPI · Docker · PostgreSQL / pgvector

SeedTracker: AI-Powered Finance & Engagement SaaS *Founder & Lead Engineer*

Church finance platform live with 8 churches in Nigeria. Built the backend services, integrated payment and WhatsApp APIs, and designed RBAC, audit trails and AI-assisted financial reporting.

Python · FastAPI · PostgreSQL · Third-party API integration

TrialForge AI & TDCT Validator *Lead Developer*

Built a rules-based validation engine running 20+ repeatable protocol-consistency checks, automating quality assurance and producing structured, auditable documentation.

Python · YAML

C# / ASP.NET & SQL DEVELOPMENT (ACADEMIC)

Built five data-driven web applications in **C# and ASP.NET** with **Microsoft SQL Server** and **MySQL** back ends: Library Management, Online Learning, Transport & Logistics, Online Shopping and Online Banking. This work gives a foundation in the Microsoft stack and relational design that complements commercial backend experience in Python.

EDUCATION & RESEARCH

MSc Information Security & Software Engineering (Distinction) — University of Greater Manchester

BEng (Hons) Software Engineering (First Class Honours) — University of Bolton

- Published peer-reviewed AI governance research in *AI and Ethics* (Springer Nature).
- Peer reviewer for the journal *Trials*.

ADDITIONAL INFORMATION

Eligible to work in the UK · Based in Sheffield · Available for hybrid working